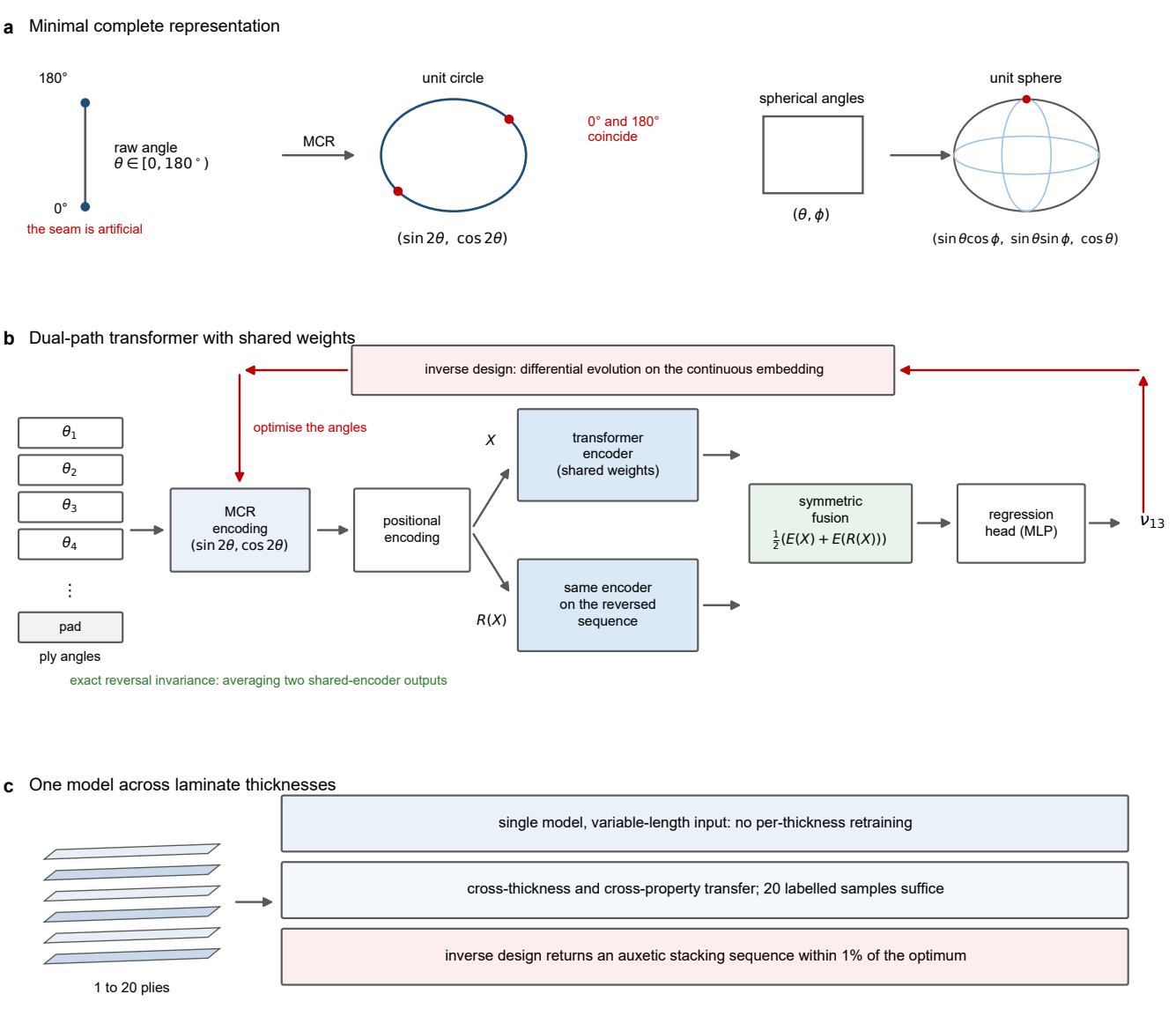
a Minimal complete representation
180°
raw angle
θ ∈ [0, 180 °)
0°
the seam is artificial
MCR
unit circle
(sin 2θ, cos 2θ)
0° and 180°
coincide
spherical angles
(θ, ϕ)
unit sphere
(sin θcos ϕ, sin θsin ϕ, cos θ)
b Dual-path transformer with shared weights
inverse design: differential evolution on the continuous embedding
optimise the angles
θ1
θ2
θ3
θ4
pad
ply angles
MCR
encoding
(sin 2θ, cos 2θ)
positional
encoding
X
R(X)
transformer
encoder
(shared weights)
same encoder
on the reversed
sequence
symmetric
fusion
½(E(X) + E(R(X)))
regression
head (MLP)
ν13
exact reversal invariance: averaging two shared-encoder outputs
c One model across laminate thicknesses
1 to 20 plies
single model, variable-length input: no per-thickness retraining
cross-thickness and cross-property transfer; 20 labelled samples suffice
inverse design returns an auxetic stacking sequence within 1% of the optimum